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1. Prove by induction that

$$n^5 - n$$

is divisible by 5 for all positive integers n

[5]

2. Prove by induction that

$$9^n - 8n - 1$$

is divisible by 64 for all positive integers $n \in \mathbb{Z}^+$

[7]

3. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$13^n + 3 \cdot 4^n + 11$$

is divisible by 12

[7]

4. Prove by mathematical induction that, for all non-negative integers n ,

$$2^n + 2 \cdot 3^{2n} + 4^{2n+1}$$

is divisible by 7

[6]

5. For each positive integer n , define

$$u_n = 5^{2n} - 2^n$$

Use proof by induction to show that 23 is a factor of u_n for all $n \in \mathbb{Z}^+$

[6]

6. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$7^n - 6n - 1$$

is divisible by 36

[6]

7. The function f is defined by

$$f(n) = 9^n - 8n + 63$$

Prove by induction that $f(n)$ is divisible by 64 for $n \geq 1$

[6]

8. Prove by mathematical induction that

$$7^{2n} + 8 \times 13^n - 9$$

is divisible by 36 for all positive integers n

[6]